

ECE 3337: Final Exam

Fall 2025

Last name of student	Student ID number
First name of student	Email address

Do This First:

1. Make sure that you have all the pages of the exam book.
2. Fill in the above information.
3. Turn off and put away all electronic gadgets (laptop, tablet, smartphone, calculator, smartwatch, wearable, ...)

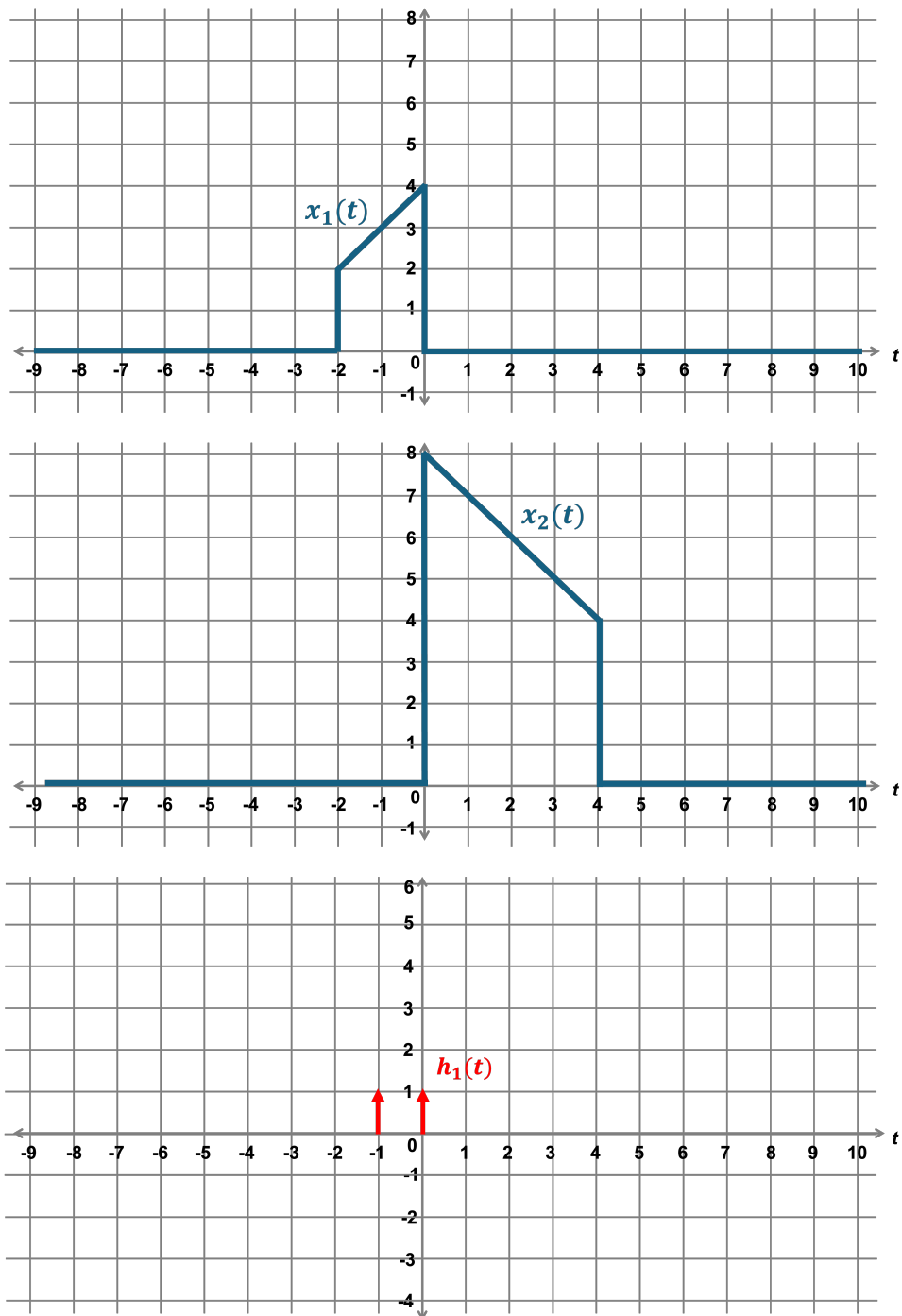
Exam Rules:

1. This is a 2-hour exam.
2. The syllabus includes all material covered in class.
3. You are allowed to bring two 8.5" x 11" handwritten crib sheets (double-sided).
4. If you need additional space to answer a question, continue on the back page(s).
5. You are not allowed to interact with any other student.

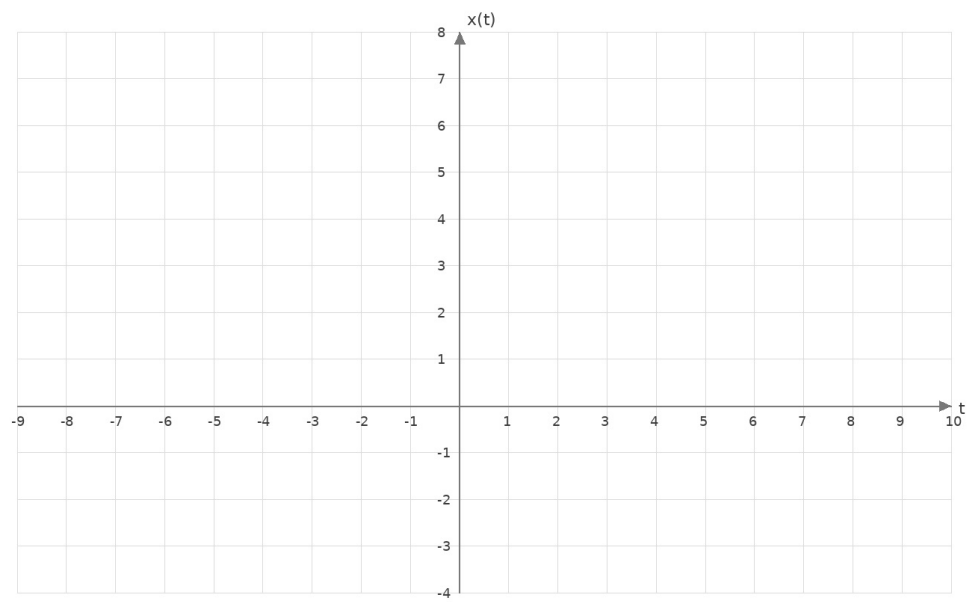
Q1	Q2	Q3	Q4	Q5	Q6	Total
15	10	10	20	25	20	100

1. (15 points) Waveforms and Convolution

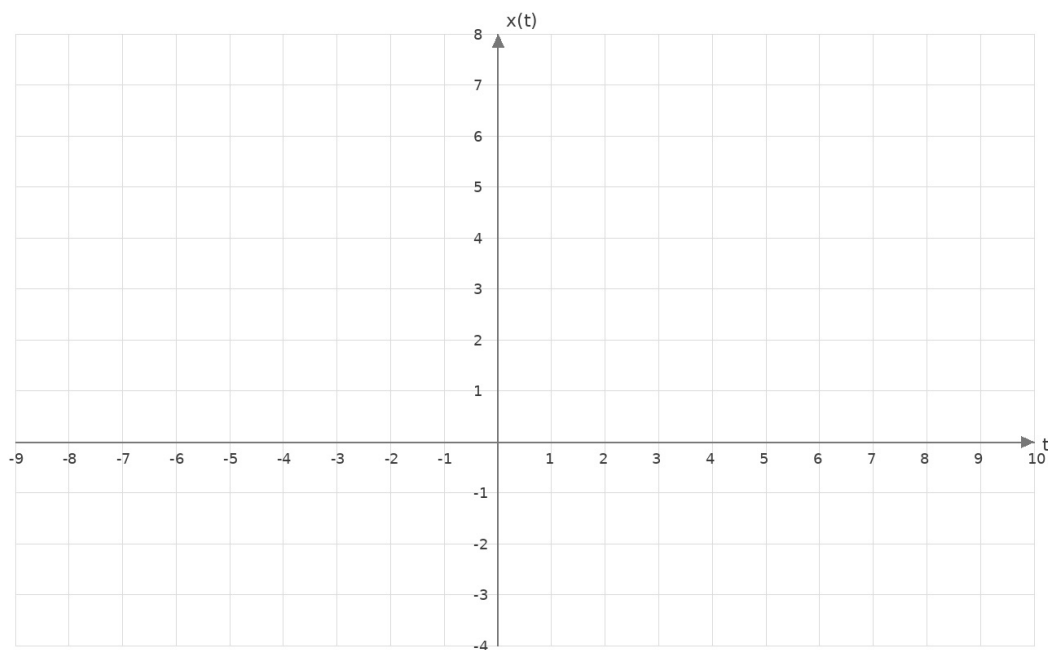
The signals $x_1(t)$, $x_2(t)$, and $h_1(t)$ are as graphed below.



- a. (7 points) Write the signal $x_2(t)$ in terms of $x_1(t)$. Hint: $x_2(t)$ is a transformed version of $x_1(t)$.

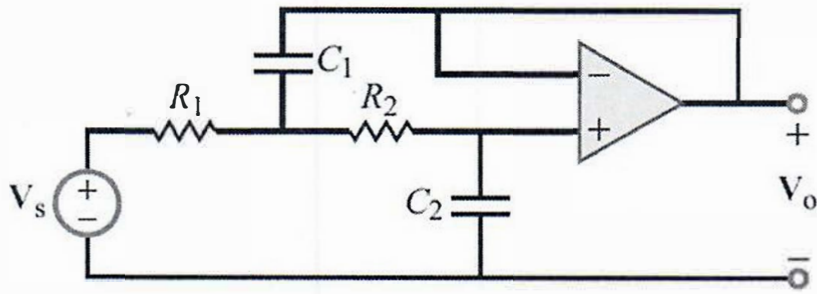


- b. (8 points) Graph the output of the convolution: $y_1(t) = x_1(t) * h_1(t)$, where $*$ represents convolution. Note that the heights of the delta function arrows indicate their amplitudes. Your final answer must be graphed below, and this will be the only answer graded.



2. (10 points) Linear Time-invariant Systems

Consider the following op-amp circuit:



a. (7 points) Derive an expression for the magnitude of the transfer function $|\mathbf{H}(\omega)|$

b. (3 points) What type of filter does this circuit represent? Explain your answer.

3. (10 points) Laplace Transform & Its Inverse

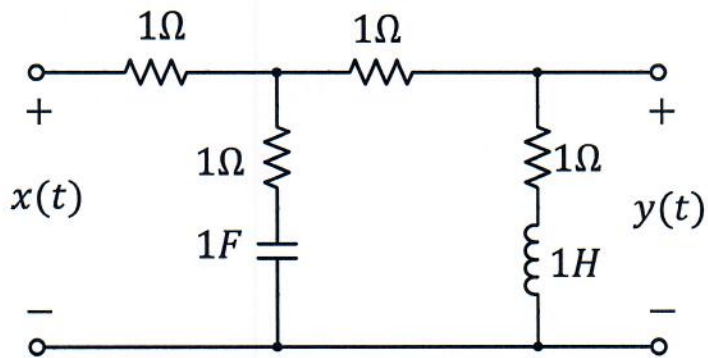
- a. (5 points) Determine the Laplace transform $\mathbf{X(s)}$ of the signal:

$$x(t) = e^{-j7t}u(t - 1)$$

- b. (5 points) Determine the inverse Laplace Transform of $\mathbf{H(s)} = \frac{s^2+10s+18}{s^2+7s+12}$.

4. (20 points) Applications of the Laplace Transform

Consider the following circuit the input $x(t) = 1V$ for $t < 0$, and $x(t) = (1 - e^{-t+1})u(t - 1)$ for $t > 0$.



- a. (2 points) Sketch the Laplace-domain version of the given circuit with the initial conditions indicated (use the back sheet for your work as needed and write the final answer below).
- b. (3 points) Write the Laplace transform of the input for $t \geq 0$ (If needed, use the back sheet for your work and write the final answer below).

- c. (5 points) Calculate the Laplace-domain output $Y(s)$ for $t \geq 0$. (If needed, use the back sheet for your work and write the final answer below). Your answer should be in the form of a ratio of two polynomials.

- d. An LTI system has the following differential equation description, with initial conditions $x(0^-) = 1$, $y(0^-) = 1$, $x'(0^-) = 1$, $y'(0^-) = 1$, $y''(0^-) = 1$, and input $x(t) = e^{-t}u(t)$:

$$\frac{d^2y}{dt^2} + 10\frac{dy}{dt} + 24y = 3\frac{dx}{dt} - x$$

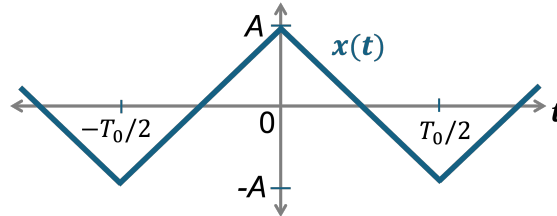
- (i) (3 points) Identify the modes of this system.

- (ii) (7 points) Calculate the natural response of this system in the Laplace domain. (If needed, use the back sheet for your work and write the final answer below).

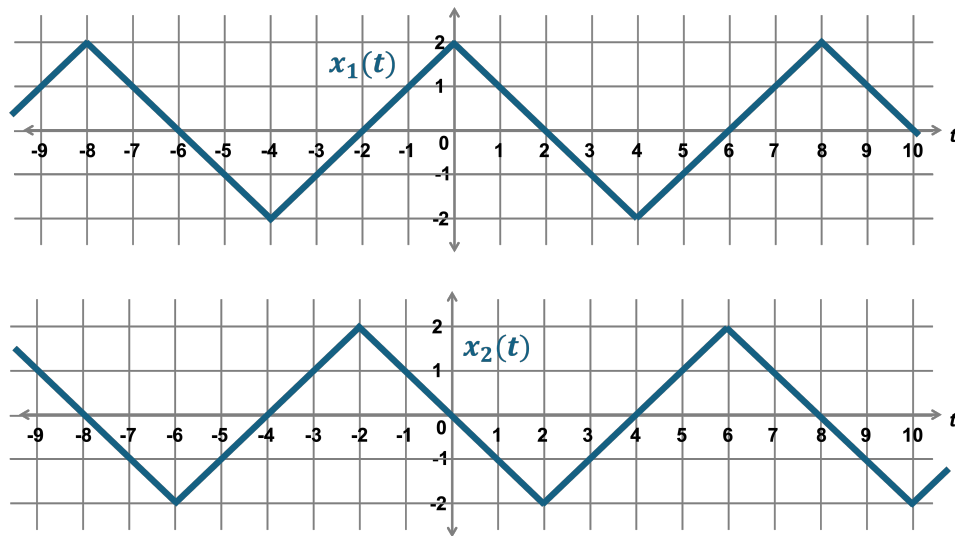
5. (25 points) Fourier Series and Transform

a. (12 points). The Fourier series of the periodic signal $x(t)$ shown below is

$$x(t) = \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty} \frac{8A}{n^2\pi^2} \cos\left(\frac{2n\pi t}{T_0}\right)$$



The signals $x_1(t)$ and $x_2(t)$ are shown below



(i) (2 point) What is the time period of the signal $x_1(t)$?

(2 point) What is the fundamental angular frequency of $x_1(t)$?

(ii) (3 points) Find the Fourier Series of the signal $x_1(t)$ in cosine/sine form.

- (iii) (5 points) Find the Fourier Series of the signal $x_2(t)$ in amplitude-phase form. i.e.

$$x_2(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \phi_n)$$

Find the values of

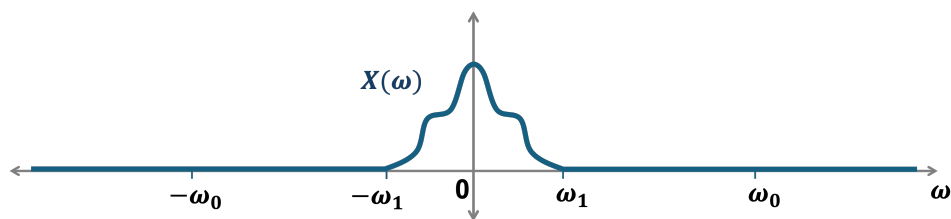
$$c_0 = \underline{\hspace{2cm}}$$

$$c_n = \underline{\hspace{2cm}}$$

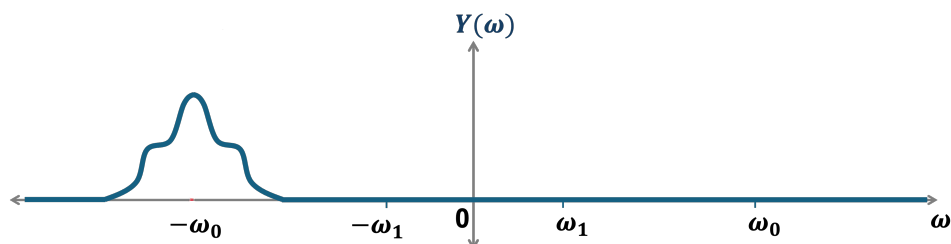
$$\phi_n = \underline{\hspace{2cm}}$$

- b. (8 points) Find the Fourier Transform of the signal $x_3(t) = 2 \cos(t) [u(t) - u(t - 1)]$.

- c. (5 points) The Fourier transform of signals $x(t)$ and $y(t)$ are $X(\omega)$ and $Y(\omega)$ respectively. $X(\omega)$ is graphed below.



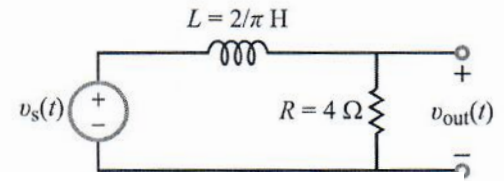
If $y(t) = x(t)e^{-j\omega_0 t}$, please graph $Y(\omega)$ in the space below. Explain your reasoning.



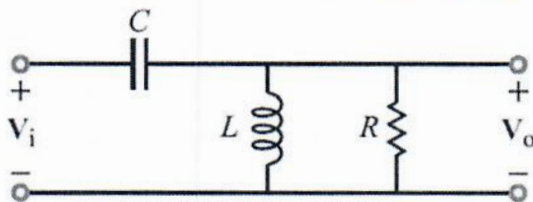
6. (20 points) Applications of Fourier Methods

- a. (7 points) Calculate the output $v_{out}(t)$ generated by the following RL circuit when the input signal is a triangular waveform. The Fourier series for the triangular waveform is given below.

$$x(t) = 10 - \frac{20}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi}{2} t\right).$$



- b. (6 points) What type of filter does the following circuit implement? If $L = 0.001H$ and $C = 10\mu F$, calculate the center frequency ω_0 and the value of R that will achieve a quality factor $Q = 10$.



- c. (7 points) A signal $x(t)$ consists of frequency in the range -10 Hz to 10 Hz. It is sampled at 12 samples/s. What range of frequencies in the signal spectrum can still be recovered from its samples without distortion? Assume that a perfect low pass anti-aliasing filter is available.